

LECTURE 5

DATA ANALYSIS

PREPARING DATA





LECTURE CONTENT

- What is Data preparation?
- Cleaning text data
- Preparing date data
- Importing data
- Find & Replace
- Removing duplicates

What is data preparation?

Data preparation is the process of gathering, combining, structuring and organizing data so it can be used in business intelligence (BI), analytics and data visualization applications. The components of data preparation include data preprocessing, profiling, cleansing, validation and transformation. It often also involves pulling together data from different internal systems and external sources.

WHAT ARE THE CHALLENGES OF DATA PREPARATION?

Data preparation is inherently complicated. Data sets pulled together from different source systems are highly likely to have numerous data quality, accuracy and consistency issues to resolve. The data also must be manipulated to make it usable, and irrelevant data needs to be weeded out.

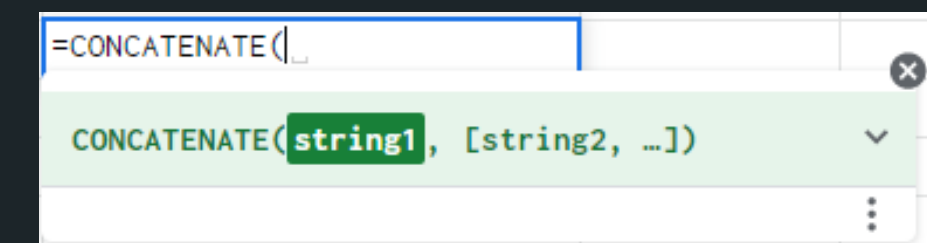
The 80/20 rule is often applied to analytics applications, with about 80% of the work said to be devoted to collecting and preparing data and only 20% to analyzing it.

CLEANING TEXT DATA

CONCATENATING TEXT STRINGS

CONCATENATE function is used to join up to 30 text items and return the result as a text.

Syntax:



Alternative to Concatenate formula



E2	fx	=CONCATENATE(A2:C2)			
	A	B	C	D	E
1	Fullname				DudkoGeorgeAdamovich x
2	Dudko	George	Adamovich		=CONCATENATE(A2:C2)
3	Kalinin	Irina	Ilyevna		
4	Mukoseeva	Valeria	Ignatievna		
5	Susnin	Nikita	Kazimirovich		
6	Trufanova	Evgeniya	Fomevna		
7	Shulgina	Faina	Rodionovna		
8	Yuferev	Charles	Potapovich		

E2	fx	=A2 & B2 & C2			
	A	B	C	D	E
1	Fullname				DudkoGeorgeAdamovich x
2	Dudko	George	Adamovich		=A2 & B2 & C2
3	Kalinin	Irina	Ilyevna		
4	Mukoseeva	Valeria	Ignatievna		
5	Susnin	Nikita	Kazimirovich		
6	Trufanova	Evgeniya	Fomevna		
7	Shulgina	Faina	Rodionovna		
8	Yuferev	Charles	Potapovich		

CONCATENATE IMPROVED

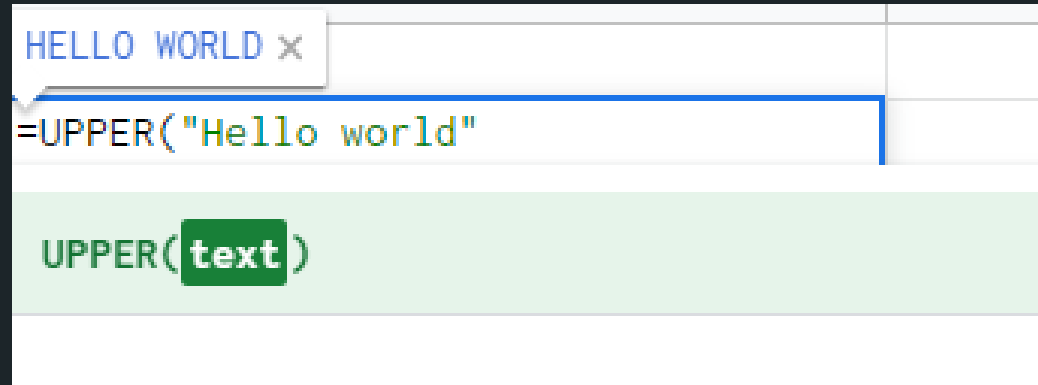
E2	fx	=CONCATENATE(A2," ", B2," ", C2,					
	A	B	C	D	E	F	G
1	Fullname				Dudko George Adamovich x		
2	Dudko	George	Adamovich		=CONCATENATE(A2," ", B2," ", C2,		
3	Kalinin	Irina	Ilyevna		CONCATENATE(string1, [string2, ...])		
4	Mukoseeva	Valeria	Ignatievna				
5	Susnin	Nikita	Kazimirovich				
6	Trufanova	Evgeniya	Fomevna		Trufanova Evgeniya Fomevna		

Space between arguments

E2	fx	=A2 & " " & B2 & " " & C2					
	A	B	C	D	E		
1	Fullname				Dudko George Adamovich x		
2	Dudko	George	Adamovich		=A2 & " " & B2 & " " & C2		
3	Kalinin	Irina	Ilyevna		Kalinin Irina Ilyevna		
4	Mukoseeva	Valeria	Ignatievna		Mukoseeva Valeria Ignatievna		
5	Susnin	Nikita	Kazimirovich		Susnin Nikita Kazimirovich		

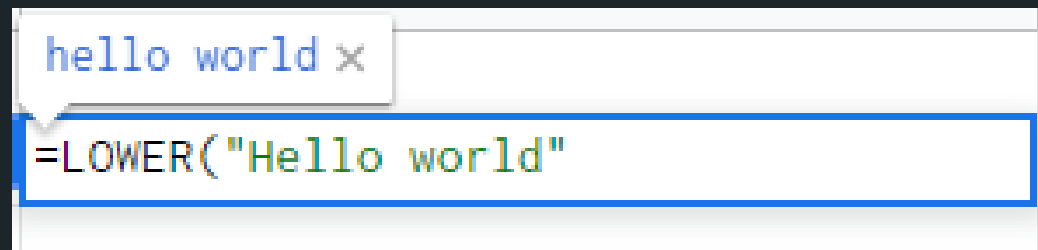
Alternative to Concatenate formula

UPPER(), LOWER() AND PROPER() FUNCTIONS

A screenshot of an Excel spreadsheet. The top row contains the text 'HELLO WORLD' in a blue font, followed by a small 'x' icon. The second row contains the formula '=UPPER("Hello world")' in a green font. The third row contains the text 'UPPER(text)' in a green font, with 'text' highlighted in a green box.

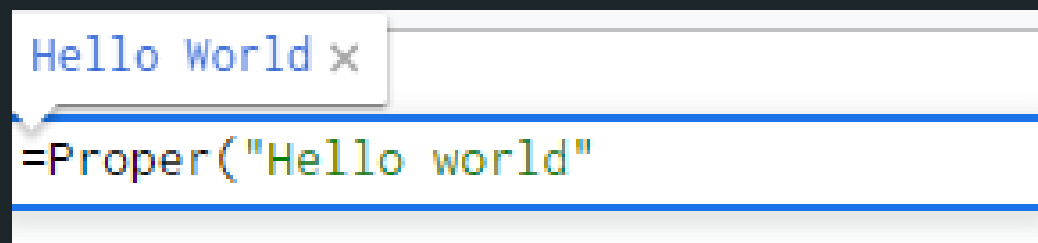
```
HELLO WORLD x
=UPPER("Hello world")
UPPER(text)
```

Converts a specified string to uppercase

A screenshot of an Excel spreadsheet. The top row contains the text 'hello world' in a blue font, followed by a small 'x' icon. The second row contains the formula '=LOWER("Hello world")' in a green font.

```
hello world x
=LOWER("Hello world")
```

Converts a specified string to lowercase

A screenshot of an Excel spreadsheet. The top row contains the text 'Hello World' in a blue font, followed by a small 'x' icon. The second row contains the formula '=Proper("Hello world")' in a green font.

```
Hello World x
=Proper("Hello world")
```

Capitalizes each word in a specified string

EXTRACTING LETTER, TEXT

	H x		
Hello World	=LEFTB(J2,1)		
	LEFTB(string, [number_of_bytes])		

Output

Syntax:

LEFTB () - Substring from beginning of specified string

RIGHTB () - Substring from the end of a specified string

	udemy x		
udemy - udemy.com	=LEFTB(J2, len("udemy"))		
coursera - coursera.com	coursera		

Output

Syntax:

LEN () - length of a string

SUBSTITUTE() FUNCTION

	udemy : udemy.com x
udemy - udemy.com	=SUBSTITUTE(J2, "-", ":")
coursera - coursera.com	coursera : coursera.com

Output

Syntax:

SUBSTITUTE () - Replaces existing text with new text in a string

=SUBSTITUTE(text, old_text, new_text)

PREPARING DATE DATA

PROJECT LENGTH

launched_at	deadline	project length
8/30/2022	9/25/2022	=M10-L10
01/12/2022	05/08/2022	116
3/11/2022	5/5/2022	55

Syntax:

deadline-launched_at

NOW(), TODAY() FUNCTIONS

deadline	now	today
9/25/2022	9.696660984	9
05/08/2022	149.696661	149
5/5/2022	152.696661	152

deadline	now	today
9/25/2022	9.69666098	=TODAY()-L15
05/08/2022	149.696661	149
5/5/2022	152.696661	152

TODAY() - current date as a date value

deadline	now	today
9/25/2022	=NOW()-L15	9
05/08/2022	149.696661	149
5/5/2022	152.696661	152

NOW() - current date and time as a date value

IMPORTING DATA

IMPORTING DATA FROM AN EXTERNAL SOURCES WITH IMPORT HTML

=IMPORTHTML("https://en.wikipedia.org/wiki/Kazakhstan", "table", 4)					January (°F)
Almaty	30/18	86/64	0/-8	33/17	
Shymkent	32/17	91/66	4/-4	39/23	
Karaganda	27/14	80/57	-8/-17	16/1	
Astana	27/15	80/59	-10/-18	14/-1	
Pavlodar	28/15	82/59	-11/-20	12/-5	
Aktobe	30/15	86/61	-8/-16	17/2	

Syntax:

=IMPORTHTML(		
IMPORTHTML(url , query, index, locale)		

IMPORTHTML() - Imports data from a table or list within an HTML page

url

The URL of the page to examine, including protocol (e.g. http://).

query

Either "list" or "table" depending on what type of structure contains the desired data.

index

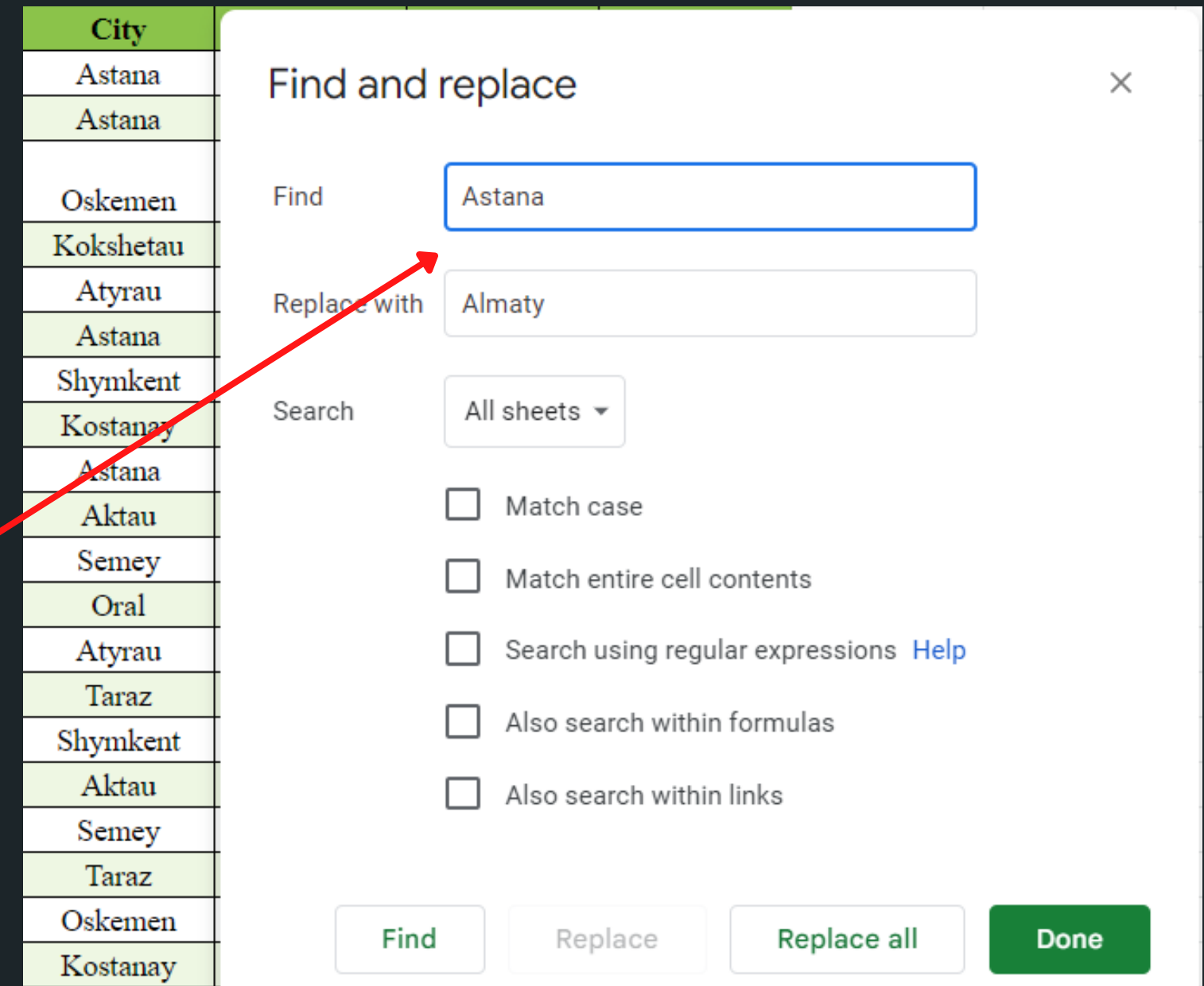
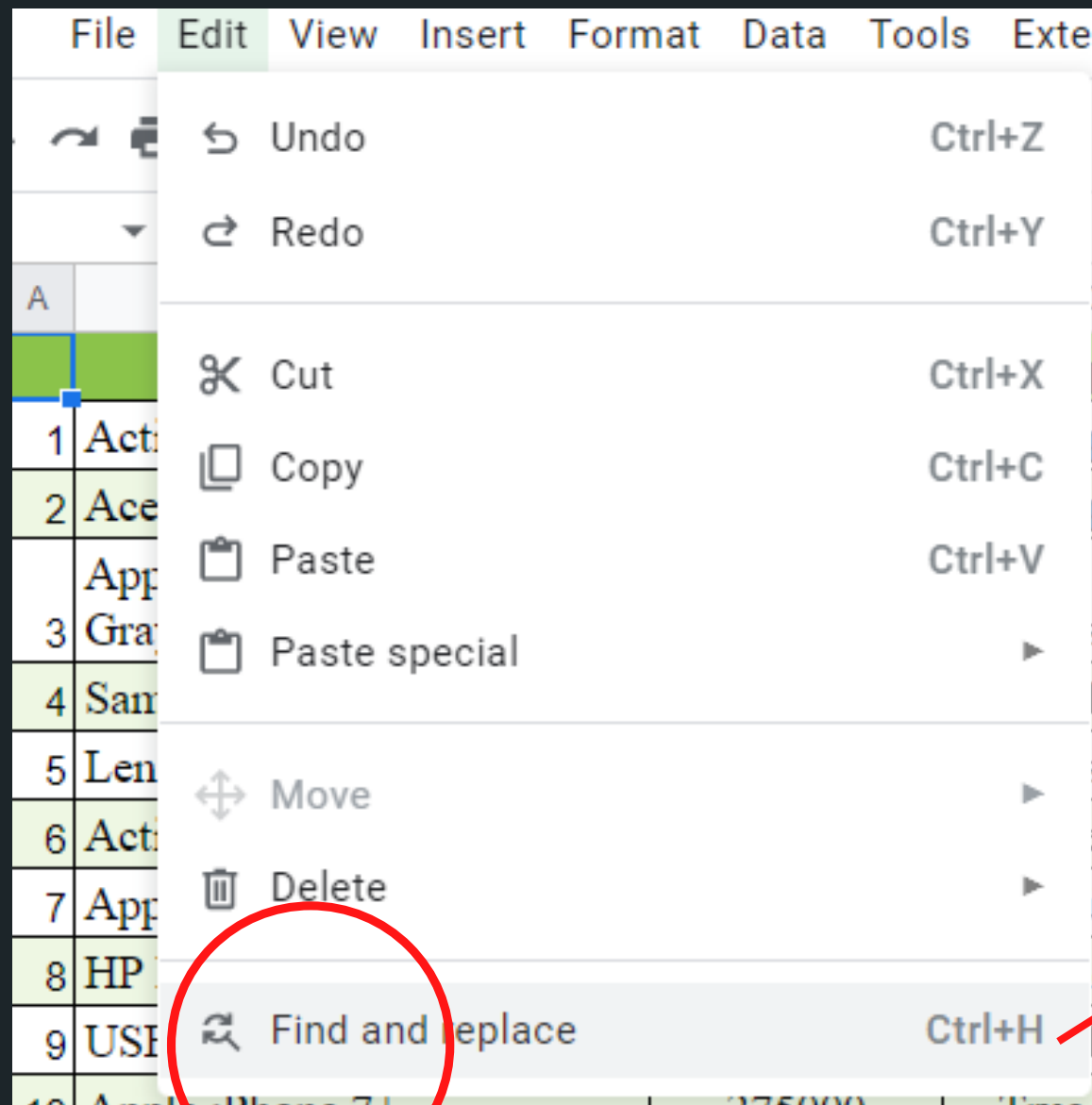
The index, starting at 1, which identifies which table or list as defined in the HTML source should be returned.

locale

A language and region locale code to use when parsing the data. If unspecified, the document locale will be used.

FIND & REPLACE

FIND & REPLACE



Find and Replace helps to find words or formats in a document and can let you replace all instances of a word or format.

REMOVE DUPLICATES

REMOVE DUPLICATES

Data Tools Extensions Help Last edit was seconds ago

Sort sheet
Sort range

Create a filter
Filter views
Add a slicer

Protect sheets and ranges
Named ranges
Named functions
Randomize range

Column stats
Data validation
Data cleanup
Split text to columns
Data connectors

New

New

Cleanup suggestions
Remove duplicates
Trim whitespace

City	Quantity	Region
Astana	5	
Astana	7	
Oskemen	5	
Kokshetau	4	
Atyrau	7	
Astana	8	
Shymkent	9	
Kostanay	2	
Astana	3	
Aktau	5	
Semey	4	
Oral	11	
Atyrau	12	
Taraz	34	
Shymkent	5	
Aktau	6	
Semey	8	
Taraz	9	
Oskemen	8	

Remove duplicates

20 rows and 1 columns selected
[Expand to A1:G28](#)

☐ Data has header row

Columns to analyze

☒ Select all
☒ Column E

Cancel Remove duplicates

City	Quantity
Astana	5
Oskemen	7
Kokshetau	5
Atyrau	4
Shymkent	7
Kostanay	8
Aktau	9
Semey	2
Oral	3
Taraz	5
	4
	11
	12
	34
	5
	6
	8
	9
	8
	14

Remove duplicates

10 duplicate rows found and removed.

10 unique rows remain.

OK

Removes duplicate lines from a list